# The Ethical Issue of Facial Recognition and Privacy

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CSC362: AI and Machine Learning

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April 2, 2026

**Abstract**

Facial recognition technology has raised significant ethical questions about security and privacy. The Integrated Joint Operation Platform in China is a large-scale surveillance program that is the subject of this paper's analysis of the ethical implications of facial recognition technology. IT investigates how these technologies may violate people's rights by enabling continuous monitoring and automated decision-making, as well as by collecting and analyzing biometric data without consent. The essay uses legal frameworks, including the Personal Information Protection Law, and finds differences between the law and its actual implementation. Recent efforts to address these issues are also discussed in this essay, including the use of new facial recognition methods that ensure user consent, data security, and transparency. The paper includes a biblical perspective on human dignity, justice, and fairness as a foundation for evaluating the ethical use of technology. Finally, while facial Recognition has significant potential to reduce crime, AI must be used carefully, aligned with ethical standards, and respectful of human rights.

**The Ethical Issue of Facial Recognition and Privacy**

Facial Recognition raises significant concerns in the context of AI and Machine Learning (ML). To recognize individual faces based on patterns and facial traits, this technology uses deep learning algorithms that analyze and compare a vast library of photos. This issue is important because governments and private companies are adopting facial recognition systems to replace ID cards or fingerprints, and often without individuals' knowledge or consent. For example, some "property management companies installing facial recognition software at the entrances to apartment complexes without the residents' permission, heightening citizens' awareness of the dangers of losing their face." (*China's Facial Recognition Concerns*, s. f.). While many individuals voluntarily use facial Recognition and similar technologies on their personal devices, non-consensual use raises ethical concerns. These concerns include questions like, "Where is this data going?" Is it secure enough? How is data management working when companies use people's faces without their consent? Is it legal? As this technology grows, we see the importance of addressing privacy rights and protecting individuals.

**Facial Recognition Technology works.**

Facial Recognition uses machine learning and neural networks to detect faces by extracting patterns and measurements of people's faces. The algorithm I will use is the Dlib CNN detector, which is a deep learning-based Maximum-Margin Object Detection (MMOD) algorithm. How the algorithm works is the following (*Face Recognition Using Artificial Intelligence*):

1. Face detection: Identify the region where a face is present.
2. Face alignment: Uses key facial landmarks such as nose, eyes, and lips.

3. Feature extraction: The deep learning model converts the landmarks into numerical vectors, called embeddings, and compares the embeddings of two faces to determine their similarity.
4. Face matching: the system compares them to a different person by using different techniques.

**Facial Recognition and its evolution**

Facial Recognition has been explored since the early 1960s. Woody Bledsoe developed a semi-automated system that required humans to use their facial features, like eyes, ears, and nose, to match them to stored images. In the 1980s, Sirovich and Kirby developed the Eigenfaces, which reduced the complexity of facial images and distinguished one face from another. In the early 2000s, the U.S. Department of Defense developed a standardized method for evaluating facial requirements. In 2010, the development of Convolutional Neural Networks (CNNs) transformed the field, enabling it to process large amounts of visual data. In 2014, Deep Face, developed by Facebook, set a benchmark of 97% accuracy in facial recognition tasks.

**Neural Networks**

Deep learning techniques use neural networks, which are a subset of machine learning. They are composed of node levels, each with an input layer, an output layer, and one or more hidden layers. Each node has a connected threshold and a weight. If a node's output exceeds the specified threshold, it becomes active and sends data to the subsequent layer of the network. If not, no data is moved to the next tier of the network. Prior to CNNs, tedious, manual feature extraction methods were used to identify objects in photos. These days, CNNs provide a more scalable approach to object identification and image classification by using linear algebraic operations, particularly matrix multiplication, to find patterns in images. However, they can be computationally demanding because they are

trained on graphical processing units (GPUs) (*What Are Convolutional Neural Networks?*, 2021).

**Case Analysis**

One example of the increase in AI and facial Recognition is seen in China. A student in Beijing reported that the school had installed a facial recognition camera in the dormitories without any sense of permission or consent from the students. At the same time, a professor at Zhou's university reported that the university added cameras in the teaching buildings without his consent to share his facial data (*China's Facial Recognition Concerns*, s. f.). This ethical issue is growing because there are no regulations governing the use of biometric data. It is unclear who is responsible for keeping biometric data safe, allowing it to be leaked and used in scams and to steal identities if not managed properly. This problem keeps growing, reaching a point where there is no freedom or privacy, whether you are on the street or in your own apartment. There are about 350 million surveillance cameras in China as of 2018; this number is forecast to keep growing as new and more advanced facial recognition technology is implemented.

In China, there is an algorithm referred to as "China's Algorithm of Repression". In a Human Rights Watch report, we can see that in Xinjiang, China, the government built an AI-powered mass surveillance system. This app collects information from individuals connected to IJOP (China's system). This system uses facial Recognition and Wi-Fi sensors to collect data. Some of the information this application collects includes name, ID number, phone usage, car details, electricity usage, religion, and social relationships. The AI's purpose is to flag in the system behaviors that are not usual; some examples Human Rights Watch emphasizes include using VPNs, WhatsApp, traveling outside the area, and having relatives abroad. All these behaviors are legal, but the AI might flag them as "suspicious." Whenever there is suspicious activity, the app sends an alert to the police through the app, and the police

can decide to investigate. Some of these cases end up in arrests. Another tool this AI has is facial detection to check whether a photo matches an ID through cross-referencing, including recognizing people from both national and foreign embassies. People are being recorded and analyzed every minute of their daily lives; if they behave outside their "routine," the system could flag it as "suspicious."

**Negative instances of Facial Recognition**

China's app raises many ethical concerns, particularly the IJOP; for instance, a main concern is the lack of consent to share personal and biometric information. Most people are unaware that their personal data is being collected and do not have the opportunity to agree or deny. According to the Personal Information Protection Law, Article 29 says, "Individual consent should be obtained for processing sensitive personal information. Where laws and administrative regulations provide that the processing of sensitive personal information shall be subject to written consent, such provisions shall prevail" (Law, 2022b). The law seeks to protect people's biometric information. However, extensive monitoring is conducted without each person's express consent, raising concerns about whether such practices comply with the law, particularly when applied to an entire population in public areas.

The application also violates the principle of "innocent until proven guilty," and basic human activities like traveling outside the country or changing your routine can be flagged as suspicious rather than reflecting growth or change as a person. These behaviors are normal and do not mean wrongdoing. However, the system can interpret them as potential threats, leading to investigations or even arrest. This approach changes the situation for individuals trying to prove their innocence when conflicting fundamental ethical and legal principles are at stake.

Additionally, the IJIOP system automatically determines whether the behavior is normal or abnormal. Even if it is harmless and there is no crime, the AI could still flag it. AI

systems do not fully understand human interacts and complexity; the lack of context or intent plays an important role in why humans act the way they do. As a result, harmless actions can be interpreted as wrongdoings and viewed as suspicious, leading to severe consequences or investigations. Article 24 of the PIPL states that individuals have the right to understand the decisions made by automated processes. However, the IJOP system offers little transparency or the opportunity for individuals to contest these decisions.

Finally, the system also collects private information, such as relationships, personal habits, and daily activities, which has raised privacy concerns. Article 28 of the PIPL states that "personal information processors can process sensitive personal information only when they have a specific purpose and sufficient necessity, and take strict protective measures" (Law, 2022a). However, the extent and depth of data collection in these public places raise questions about their necessity and proportionality. Monitoring individuals' movements and behaviors daily can be seen as invasive surveillance rather than a security effort, challenging the rationale for the government's use of this technology.

**Positive efforts to mitigate Facial Recognition concerns**

China has also sought to implement new rules and regulations governing facial Recognition. To safeguard personal information and prevent misused, the Security Management Measures for the Application of Facial Recognition Technology were established in 2025. Starting with storage, it prohibits external transfers unless the data subject gives separate authorization and relevant laws and regulations permit the transfer. Before processing the data, the Privacy Impact Assessment ("PIA") must also contact the subject.

The measure also adds a limitation on public places where facial recognition devices can be installed; however, it must be subject to the data handler establishing the necessity for maintaining public security. The data handler shall reasonably determine the area for facial

information collection and display prominent warning signs(*China Releases New Rules Regarding the Use of Facial Recognition Technology*, s. f.). In addition, the data handler should not rely solely on facial Recognition as the verification method if another technology can achieve the same purpose or meet equivalent business requirements. Companies with more than 100,000 individuals that use facial recognition technology should file with the competent Cyberspace authority at the provincial level or higher within 30 business days upon reaching that threshold (*China Releases New Rules Regarding the Use of Facial Recognition Technology*, s. f.).

In addition to these regulations, China aims to address concerns about data security and unauthorized surveillance. To protect people's privacy, businesses should avoid using facial Recognition in sensitive areas such as hotels, public restrooms, and private places. They should also explain the reasons and purposes for using biometric data. Additionally, companies must be open about how they use and collect biometric data, including how long it takes and how it can affect specific people. These notices must be simple to understand, and the company is obligated by law to notify people if the data it uses changes. Prior to collection, specific parental consent is required for special protections for kids' facial data. Companies must implement additional security measures and follow specific rules on data use, retention, and transfer to protect their data and prevent misuse.

Finally, the measures introduce more security and risk management; there are strict regulations on data encryption, access controls, regular security audits to identify vulnerabilities, risk assessments, and incident response protocols to implement corrective actions.

**Facial Recognition VS Faith**

Genesis 1:27 emphasizes that as humans, “God created man in his own image, in the image of God he created him; male and female he created them” (*Genesis 1:27 NIV - So God*

*created mankind in his own - Bible Gateway*, s. f.). This idea shows that every human should be treated with respect, autonomy, and morality. When we talk about Facial Recognition, this raises concerns about the systems monitoring individuals without consent or reducing people to data points. Continuous surveillance can undermine the sense of freedom and personal dignity that come from being recognized as individuals created with purpose and value. From this perspective, the use of AI needs to be carefully considered to ensure it does not dehumanize or exploit individuals.

Additionally, the Bible highlights the importance of fairness and justice. Proverbs 21:3 states, "To do righteousness and justice is more acceptable to the LORD than sacrifice" (*Proverbs 21:3 NIV - To do what is right and just is more - Bible Gateway*, s. f.). These principles challenge the use of facial Recognition because the system may unfairly target or flag individuals due to bias. Systems with large surveillance platforms risk treating people suspiciously without evidence, which conflicts with biblical principles. The ethical use of technology should be prioritized, with individuals protected, rather than creating and reinforcing systems that may lead to unjust outcomes.

**Personal Faith Reflection**

One of my closest experiences with biometric data occurred at the airport in Colombia, where you can skip traditional immigration by registering in a biometric system that scans your eyes. This system allows travelers to pass through security without standing in long lines and interacting with an officer. My second experience was in college, when I applied for OPT through U.S. Citizenship and Immigration Services (USCIS). As part of the process, I was required to provide a photo as biometric data. In both cases, the biometric data was reasonable, limited to a specific purpose, and did not include day or night surveillance.

From my personal faith, human dignity and freedom are essential values that guide technology. While I understand that governments are responsible for public safety, there

needs to be clear limits on the use of facial Recognition. One example will be using biometric systems to investigate cases of crime or other violations. However, recording and recognizing people in their daily lives is a complete violation of privacy; freedom will no longer be freedom, and the government will manage people's lives.

At the same time, facial Recognition is going to keep growing and will be used more often than it is now. When applied properly, AI technology can deliver great results. It might contribute to a decrease in crimes, including human trafficking, violence, and kidnappings. From a faith perspective, technology should serve humanity, not control it; instead of monitoring ordinary people, companies should direct AI to address real issues and protect vulnerable people. In this way, technology will be used ethically and can align power with the values of justice, compassion, and respect for human life.

**Conclusion and Solutions**

To conclude, Facial recognition technology is a powerful and sensitive topic that can get out of hand if not used properly. The essay discusses the advantages and disadvantages of facial recognition technology. For example, when used appropriately, it could reduce crime such as violence, theft, and human trafficking. However, systems like the IJOP demonstrate how technology can be abused to monitor people, restricting their freedom and leading to unfair choices motivated by prejudice. In the absence of appropriate regulations, facial recognition technology may become a control mechanism rather than a tool for protection, posing grave ethical questions about privacy, human rights, and fairness.

To address some concerns, first, biometric data should be stored in a highly secure location, using encryption and strict access controls to prevent data breaches. Government and organizations should ensure data collection is limited to what is necessary and obtain consent from the subjects. Additionally, protections should be put in place for minors, including parental consent and enhanced security for their biometric data. In public spaces,

people need to be aware that they are being filmed, like in stores or malls. This will allow people to trust that, in an emergency, they can recall the events.

From a faith perspective, these solutions are aligned with biblical principles that ensure justice and fairness. Technology should be used to serve and protect people, not control us. Respecting privacy reflects human dignity while ensuring fairness in AI systems. By always putting people first and ensuring everyone is safe, society can achieve a balance in which facial Recognition contributes to safety and progress without compromising fundamental human rights and freedoms.

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